Functions

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SOURCE: Spivak, *Calculus*, 4th Edition, Chapter 3

DEPENDENCIES

- `foundations/01-field-axioms` - Field axioms — arithmetic on function values
- `foundations/02-order-axioms` - Order axioms — inequality on $\mathbb{R}$

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1 The Definition of a Function

We now introduce four definitions that formalize the notion of a function.
Vi introduserer nå fire definisjoner som formaliserer funksjonsbegrepet.
我々は今、関数の概念を形式化する四つの定義を導入する。

Definition 1.1 (Function)**.**

Definition 1.2 (Domain)**.**

Definition 1.3 (Function Value)**.**

Definition 1.4 (Equality of Functions)**.**

2 Derived Definitions

We now define several specific functions and derived operations on functions.
Vi definerer nå flere spesifikke funksjoner og avledede operasjoner på funksjoner.
ここで、複数の具体的な関数と関数に対する派生演算を定義する。

2.1 Pointwise Arithmetic

Definition 2.1 (Sum)**.**

Definition 2.2 (Difference)**.**

Definition 2.3 (Product)**.**

Definition 2.4 (Quotient)**.**

2.2 Composition

Definition 2.5 (Composition)**.**

2.3 Specific Functions

Definition 2.6 (Identity Function)**.**

Definition 2.7 (Characteristic Function)**.**

3 Consequences

We now derive three consequences of composition.
Vi utleder nå tre konsekvenser av komposisjon.
以下では合成の三つの帰結を導く。

Proposition 3.1 (Associativity of Composition)**.**

Proof. ■

Proposition 3.2 (Identity is Neutral for Composition)**.**

Proof. ■

Proposition 3.3 (Non-Commutativity of Composition)**.**

Proof. ■